A Study of Compound Action Potentials in Current-Coupled Tracts of Identical Axons

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Abstract

In this paper, the authors investigate compound action potentials formed when the underlying tract's axons have current-mediated coupling amongst themselves. The key finding of the paper is that, for the case of biophysically identical axons, the compound action potential is governed by a Hodgkin-Huxley like equation itself. This leads to the intriguing view that a nerve is 'fractal' in nature, with the compounded signals and their components governed by similar equations.

1 Introduction

The compound action potential is a more relevant quantity than the single-fiber action potential for the neurophysiologist who does not often have access to individual fiber voltages. Thus its detailed study is of importance. In this paper we take an analytical approach and develop the resulting compounded signal equation. Some of the questions we are interested in answering include: Can a nerve be treated like an axon mathematically when compounded? If so, can there be cross-nerve current- or field-mediated coupling, say, when there is minimal insulation on the nerve? Further, if a nerve is like an axon, then all synapses emerging from a given nerve may not only be synchronized, but they may act as one unified, distributed synapse. Thus, can nerve formation be seen as a case of the organism trying to accomplish a task requiring amplification of the control signal? In this paper, we don't directly address all these questions, but our work illumines some of them.

Prior work on compound action potentials includes (Wijesinghe *et al.*, 1991), (Stegeman & De Weerd, 1982) as well as (Kim & Kim, 1976). In this literature review we will focus on (Wijesinghe *et al.*, 1991) wherein the authors study the compound action potential and develop a model for its description on the basis of single fiber action potentials. They study the effect of the myelin sheath and also the effects of temperature and fiber diameter on the compound action potential. Their forward model consists of summations of the single fiber action signals where each signal is time-shifted by the corresponding delay τ_j between the stimulus time and the arrival time at the recording electrode of that fiber.

They present a block diagram for the compound action current (CAC) and present an algorithm for computing the compound action signal. They present time-domain simulation results. They also study variation of the peak-to-peak amplitude of anisotropy of the nerve bundle. Their principal result is the relation between the conduction velocity distribution and the specific compound action potential studied.

This paper is organized as follows. In Section 2 we describe the model being studied. In Section 3 we compare the compounded signal to the action of the vector ephaptic perceptron. Finally, we conclude in Section 4.

Notation

The notation used in this paper is succinctly summarized in Table 1.

Table 1: Table of Notation

S. No.	Symbol	Meaning
1	C	fiber capacitance per unit length
2	α, β	constants related to intra- and extra-cellular resistivity, ...
		... summarizing the constants in (Reutskiy *et al.*, 2003)
3	g	constant related to conductance of myelin
4	V_1, V_2	transmembrane potential of axon 1 and axon 2
5	τ_1, τ_2	time delay between stimulus onset and the ...
		... action potential initiation time on fibers 1 and 2

2 The Model and Its Study

As per (Wijesinghe *et al.*, 1991), under the assumption of superposition, the compounded action potential (CAP) is given in terms of time-shifted versions of the single fiber action potentials:

$$CAP(x,t) = \sum_{j=1}^{N} V_j(v_j, t - \tau_j) \tag{1}$$

where τ_j is the time-delay between the stimulus onset and the arrival time at the j-th fiber, x is the propagation distance, and the variables v_j represent the conduction velocities of the various fibers. In (Wijesinghe *et al.*, 1991), the v_j are distinct; in our case, they are identical. Nevertheless a variable propagation delay arises due to the geometry of the situation (Chawla *et al.*, 2019). More specifically, as in (Chawla *et al.*, 2021), these delays are related to the action potential initiation times at the current-coupled axons.

Consider the (Reutskiy *et al.*, 2003) equations for two coupled axons (that is, $N = 2$), both being identical biophysically[1]:

[1]The velocity variables are omitted, being identical.

$$C\frac{\partial V_1}{\partial t} = \alpha\frac{\partial^2 V_1(t)}{\partial x^2} + \beta\frac{\partial^2 V_2(t)}{\partial x^2} - gV_1 \tag{2}$$

and

$$C\frac{\partial V_2}{\partial t} = \alpha\frac{\partial^2 V_2(t)}{\partial x^2} + \beta\frac{\partial^2 V_1(t)}{\partial x^2} - gV_2 \tag{3}$$

Next, let $V = V_1(t-\tau_1) + V_2(t-\tau_2)$ be the compounded signal. Then,

$$C\frac{\partial V(t)}{\partial t} = (\alpha+\beta)\frac{\partial^2 V(t)}{\partial x^2} - gV(t) \tag{4}$$

is the equation for the compounded signal. This resembles the Hodgkin-Huxley equation for a single fiber. Thus, the compounded signal's propagation is similar to that of the single fiber signal, in the case considered (identical current-coupled fibers). Thus, in this paper we have identified a type of *self-similarity* in the nervous system.

3 Comparison With the Vector Ephaptic Perceptron

Can the compounded signal inspire a new type of perceptron (Rosenblatt, 1958)? Equation (4) apparently resembles the equation for a vector ephaptic perceptron (Equation (6) in Chawla (2022b)) and in this section we explore similarities and differences with the development in that paper.

We immediately note that β plays the role of λ in that paper. When β is set to zero, Equations (2) and (3) uncouple. They then resemble the single neuron equations (1) and (2) of (Chawla, 2022b). Further, with a nonzero β, we can express Equations (2) and (3) in the vector format, yielding similarity with Equation (6) of (Chawla, 2022b).

There are thus no exact parallels of Equation (4) in that paper. However, it is not entirely inconceivable, that one could have vector ephaptic perceptrons composed of compounded neural signals as parts. This is an investigation left for future work.

4 Future Work and Conclusion

To summarize, in this paper we found that when the underlying current-coupled axons are biophysically identical, then the compounded signal is like a Hodgkin-Huxley signal. This led to a remark that the nervous system may have some self-similarity or 'fractal' character to it (Mandelbrot, 1990), which would be interesting to explore further. We also devoted a section to possible implications for vector ephaptic perceptrons introduced in a previous study.

A limitation of our work is that we didn't explore the case of non-identical fibers arranged in a nerve. Furthermore, we didn't explore the case of sensorimotor nerves wherein individual fibers carry signals in two opposing directions.

In future work, we can also consider simulations of the propagation of ephaptic CAPs (eCAPs) under focal demyelination conditions. Another investigation that may yield novel insight is the impact of single fiber internodal lengths on the conduction velocity of eCAPs. However, the most important aspect of the present work is that it illumines future studies of quantum effects for entire nerves, in the context of (Chawla, 2022a) and (Chawla & Morgera, 2022).

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References

Chawla, Aman. 2022a. A gedanken experiment on the brain as a quantum computer. *Researchgate preprint.*

Chawla, Aman. 2022b. A mathematical model for the vector ephaptic perceptron. *Zenodo.*

Chawla, Aman, & Morgera, Salvatore Domenic. 2022. *Mechanism of universal quantum computation in the brain.*

Chawla, Aman, Morgera, Salvatore, & Snider, Arthur. 2019. On axon interaction and its role in neurological networks. *Ieee/acm transactions on computational biology and bioinformatics.*

Chawla, Aman, Morgera, Salvatore D., & Snider, Arthur D. 2021. Fields, geometry, and their impact on axon interaction. *Journal of applied mathematics and physics*, **9**(4), 751–778.

Kim, Yong I, & Kim, M. 1976. Control systems model of the spontaneous and light-evoked patterns of the compound action potentials in the eye of aplysia. *Biological cybernetics*, **23**(4), 203–209.

Mandelbrot, Benoit B. 1990. Negative fractal dimensions and multifractals. *Physica a: Statistical mechanics and its applications*, **163**(1), 306–315.

Reutskiy, Sergiy, Rossoni, Enrico, & Tirozzi, Brunello. 2003. Conduction in bundles of demyelinated nerve fibers: computer simulation. *Biological cybernetics*, **89**(6), 439–448.

Rosenblatt, Frank. 1958. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological review*, **65**(6), 386.

Stegeman, DF, & De Weerd, JPC. 1982. Modelling compound action potentials of peripheral nerves in situ. i. model description; evidence for a non-linear relation between fibre diameter and velocity. *Electroencephalography and clinical neurophysiology*, **54**(4), 436–448.

Wijesinghe, Ranjith S, Gielen, Frans LH, & Wikswo, John P. 1991. A model for compound action potentials and currents in a nerve bundle i: The forward calculation. *Annals of biomedical engineering*, **19**(1), 43–72.